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From epidemiological question to executable pipeline

LLM-assisted Nextflow
DSL2 synthesis, on-premise
a proof of concept

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Acknowledgements

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Martinus Grady	lead architecture & agent engineering
Ligan Cai	co-author, evaluation
Zeynal Mardanli	co-author, frontend



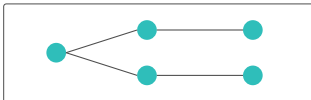
What it does

In one slide, before the rest of the talk

a question, in plain English

"I have Illumina paired-end reads of Salmonella isolates. I want to do SNP-based cluster analysis across samples. What do you suggest?"

or a canvas



Cohesive LLM

one GPU
≈ 50 s

 **cohesive-ngsmanager**

52 components
+ knowledge graph

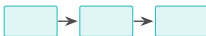
read at query time,
not trained into the model

 **genpat-it/**
cohesive-ngsmanager

an executable pipeline

pipeline.nf

the topology

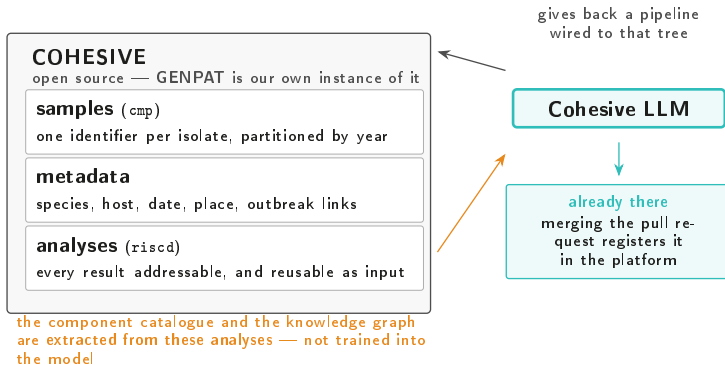


checked by Nextflow

what is missing to run

Where the pipelines live: COHESIVE

The open-source release of GENPAT, the instance IZS Teramo runs in house



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Demo

from question to validated pipeline

the whole loop 6:12 — the results 1:01 — the canvas 1:26

 watch the three parts

 or all of it, 8:38

Who it is for

Two kinds of user, and a picture both can read

1. Domain experts are demanding, but not coders.

An epidemiologist knows exactly which analysis is needed and is exacting about its scientific correctness — but cannot write DSL2, and today waits in the bioinformatics queue.

2. Experts should not start from a blank file either.

A generated skeleton, grounded in the existing catalogue and already validated, is a **starting point to edit** — not a replacement for expertise.

3. A diagram is read faster than code.

Every pipeline ships with a Mermaid flowchart: the expert checks the topology at a glance, the non-coder sees what will actually run.

Same tool, two entry points: a question in plain English, or a canvas.

Conclusions

What I would like you to take away

- **It supports the curator, it does not replace one.** The agent proposes; a person merges.
- **Constrained by design** — it composes only components already running in production, which is what makes reproducibility possible.
- **Local, and not only for the EU AI Act.** One on-premise GPU: nothing leaves the institute. The next step is to let it read the **sample metadata** as well — and that is data we cannot send anywhere. Guardrails are also easier when the output space is this narrow.
- **Portable** — nothing here is specific to GENPAT: point it at another catalogue and it works.

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Thank you

Questions?

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 [genpat-it/cohesive-llm](https://github.com/genpat-it/cohesive-llm)